

SOLUTIONS

$$1) \begin{cases} \dot{x}_1 = x_1 + 2u \\ \dot{x}_2 = -x_2 + x_1 + u \\ y = x_1 \end{cases}$$

A) Eigenvalues: $\lambda_1 = 1$ $\Rightarrow$ The system is unstable
 $\lambda_2 = -1$

Modes: $e^{\lambda_1 t} = e^t$
 $e^{\lambda_2 t} = e^{-t}$

B) a) $\frac{N_B}{R} \geq a \log_2 e = 1.442$ ($a=1$)

b) $N_B = 2 \Rightarrow R \leq \frac{2}{1.442}$

c) $N_B = 2$
 $h = \log_2 e$ $2 > \log_2 \log_2 e = \log_2 2^{\log_2 e} = 1 \Rightarrow$ THE SYSTEM IS BOUNDABLE

We apply the control law to the equation

$$\dot{y} = y + 2u \Rightarrow y_{k+1} = f y_k + g u_k = 2 y_k + 2 u_k$$

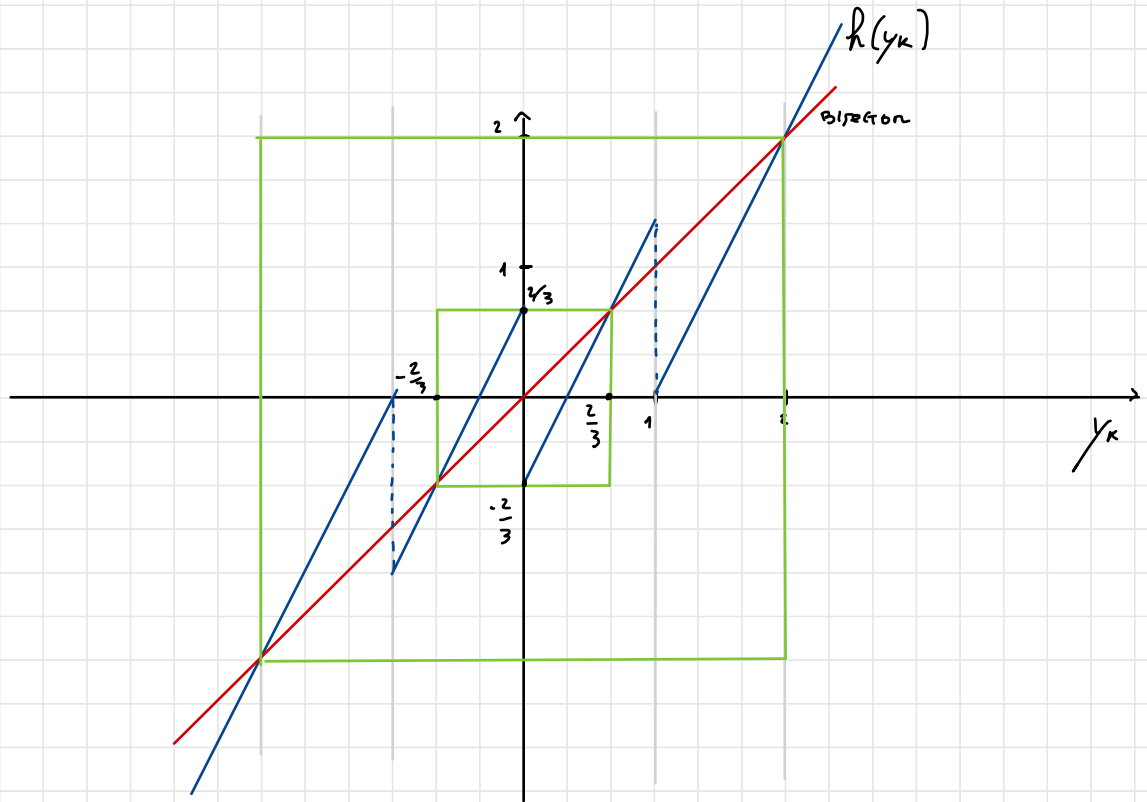
$$f = e^{aR} = 2$$

$$N = 2^{N_B} = 4$$

$$g = \frac{1}{a} (e^{aR} - 1) = \frac{2}{1} (2 - 1) = 2$$

$$u_k = \begin{cases} 1 & y_k \leq \frac{2}{1} (-1 + \frac{2}{2}) = -1 \\ 1 - \frac{2}{3} = \frac{1}{3} & -1 < y_k \leq 0 \\ -\frac{1}{3} & 0 < y_k \leq 1 \\ -1 & y_k > 1 \end{cases}$$

$$y_{n+1} = R(y_n) = \begin{cases} 2y_n + 2 & y_n \leq -1 \\ 2y_n + \frac{2}{3} & -1 < y_n \leq 0 \\ 2y_n - \frac{2}{3} & 0 < y_n \leq 1 \\ 2y_n - 2 & 1 < y_n \end{cases}$$



$$\text{IF } |y(0)| > 2 \Rightarrow |y_k| \rightarrow +\infty$$

$$\text{IF } |y(0)| = 2 \Rightarrow |y_k| = 2 \quad \forall k$$

$$\text{IF } |y(0)| < 2 \Rightarrow |y_k| < 2 \quad \forall k \Rightarrow [-2, 2] \text{ IS AN INVARIANT SET}$$

FROM THE FIGURE ABOVE, IT IS CLEAR THAT THE SYSTEM PRESENTS A FURTHER INVARIANT SET, I.E.,

$$\left[-\frac{2}{3}, \frac{2}{3}\right]$$

d). $x_1 = y$ (see point c)

$$\bullet \dot{x}_2 = -2x_2 + x_1$$

SINCE x_1 IS BOUNDED (IF $|x_1(0)| \leq 2$)
AND $\lambda_2 = -2 < 0$

$\Rightarrow x_2$ IS ALSO BOUNDED

e) The control law defined in c) is NOT the only one that allows to steer the state in a bounded set, since

$$\frac{N_0}{L} > a \log_e \rightarrow \text{NOT ON THE RATE LIMIT}$$

$\Rightarrow$ Add details and possibly an example (see Theory)

2) }
3) } See theory

